

Circumgalactic dust halos update #2

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1 Executive Summary

This is a short follow-up to the comprehensive “Circumgalactic dust halos update” document circulated in October 2023. Here are the principal updates since that time:

1. Implemented redshift-based matching for augmenting redMaGiC random catalogs with galaxy colors.
2. Fixed a bug in the generation of random (foreground) catalogs; my previous algorithm, which randomly sampled HEALPixels and then calculated the central RA and Dec of said HEALPixel, did not uniformly cover the survey area. The change led to nicer-looking results, especially for the high- z redMaGiC catalog.
3. Implemented a patch-based approach to variance estimation in correlation function calculations, as strongly recommended in `Treecorr` documentation.
4. Implemented a new dust extinction law taken from Gordon et al. 2023; the one I had been using dated from 2000 and was due for an update.
5. Started using the CSFD map recommended in Yi-Kuan Chiang 2023, as the canonical SFD98 extinction map has CIB contamination, leading to $10^{-3} - 10^{-4}$ mag excesses in $E(B-V)$. This didn’t make a huge difference, but is the right thing to do.
6. Tested for systematics using Gaia star catalogs, which yielded quite good results; see Figure ??.
7. Fit extinction profiles to the assorted redMaGiC-background fits; came up with the same slope as Ménard et al. 2010.
8. Expanded analysis to galaxies in the Sloan Digital Sky Survey and GALEX-WISE-SDSS Legacy Catalog! The results aren’t as amazing as one would hope, but aren’t complete messes either.

The code and analysis improvements are covered in Section 4. Information about the new catalogs is presented in Section 3. Results of systematics testing and profile fitting are in Section 5, and next steps are covered in Section 6.

2 Refresher: optimal estimator for dust extinction and cross-correlation calculation

Update #1 provided some background for the optimal estimator $\hat{\rho}$ for dust extinction (Equation 1. The essential equations are reproduced here for convenience.

$$\hat{\rho} = \left[\frac{\frac{\partial \mathbf{M}}{\partial \rho} \mathbf{C}^{-1} (\mathbf{D} - \mathbf{M}_0)}{\frac{\partial \mathbf{M}}{\partial \rho} \mathbf{C}^{-1} \frac{\partial \mathbf{M}}{\partial \rho}} \right] \quad (1)$$

By the Cramér-Rao bound, the uncertainty on $\hat{\rho}$ is given by:

$$\sigma_{\hat{\rho}}^2 = \frac{\partial \mathbf{M}}{\partial \rho} \mathbf{C}^{-1} \frac{\partial \mathbf{M}}{\partial \rho} \quad (2)$$

As with any such analysis, the contribution of random correlations, selection biases, etc. must be taken into account. We use the following estimator for galaxy-extinction cross-correlation $\xi_{NA_V}(\theta)$:

$$\xi_{NA_V}(\theta) = D_N D_{A_V}(\theta) - D_N R_{A_V}(\theta) - R_N D_{A_V}(\theta) + R_N R_{A_V}(\theta) \quad (3)$$

$D_N D_{A_V}$ is the signal from the correlation of the real foreground galaxy and background color catalogs; $D_N R_{A_V}$ and $R_N D_{A_V}$ are the correlations of real foreground with random background catalog and random foreground with real background catalogs, respectively; and $R_N R_{A_V}$ is the correlation of the random foreground and background catalogs.

3 Catalogs Round II

In the first update, all results presented were made using the WISExSCOS photometric redshift catalog as a foreground and the Dark Energy Survey redMaGiC catalogs (Rozo et al. 2016) as backgrounds. Since that time, there have been a few minor changes to the way the foreground and background samples are constructed. First, the redshift cut for the foreground WISExSCOS catalog galaxies is now $z < 0.2$ from $z < 0.15$, and a cut of $z > 0.5$ is placed on the high-density redMaGiC sample.

For the first time in this project, I have expanded our analysis to galaxies in the Northern hemisphere. As originally proposed in the ADAP, the northern background sample is drawn from the Sloan Digital Sky Survey Data Release 18 catalog XXX ref needed XXX. The Sloan Digital Sky survey covers...

XXX more details about query used would go here XXX

The WISExSCOS catalog is all-sky and covers the SDSS area, so that was used as to construct a foreground sample for the SDSS background galaxy catalog. I also expanded to using the GALEX-WISE-SDSS Legacy Catalog (GSWLC), with physical properties for almost 660,000 galaxies at redshifts $0.01 < z < 0.3$, as a foreground.¹ Several GSWLC variants are available; this analysis uses the GSWLC-X2, which combines the deepest areas of the GSWLC-A, M and D catalogs and covers 90% of the SDSS area (Salim et al. 2016). The GSWLC-X2 also features more accurate

¹<https://salims.pages.iu.edu/gswlc/>

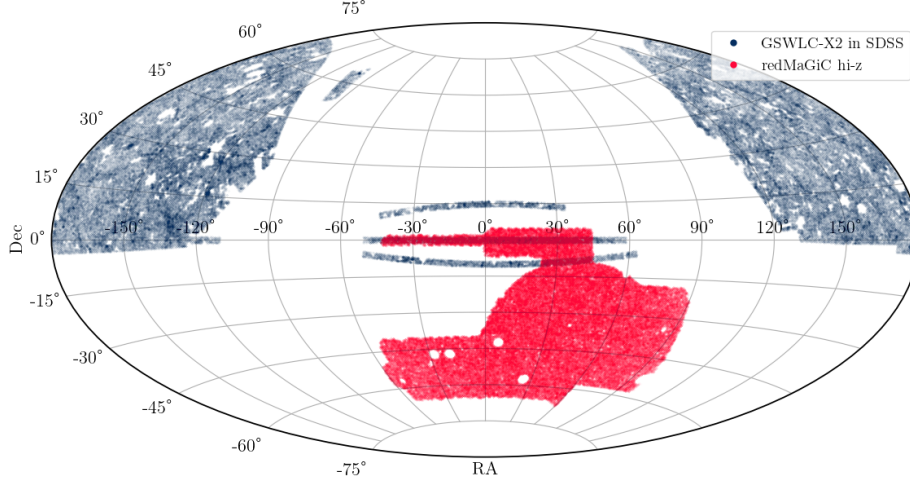


Figure 1: Sky coverage of redMaGiC high-luminosity high-redshift galaxy sample (red points) and GALEX-WISE-SDSS Legacy Catalog sources in the SDSS main galaxy sample (blue points).

star formation rates than the first generation of the catalog thanks to joint UV, optical, and mid-IR SED fitting (Salim, Boquien & Lee 2018). The full GSWLC-X2 contains 659,229 galaxies; as recommended by the authors, we restrict our analysis to the 610,518 that are included in the Sloan Digital Sky Survey main galaxy sample (`flag_mgs = 1`, see Figure 1).

Random galaxy catalogs are not provided for either GWSLC-2 or SDSS. To be able to calculate all terms in Equation 3, random galaxy catalogs for these two samples are created using the method first described in update #1 Section 4 and revisited in this update in Section 4.1.2.

Beyond new galaxy catalogs, since update #1 I have also constructed star catalogs to test for systematics in the analysis. The rationale for using star catalogs is thus: unless embedded in a star-forming region, stars do not have extended halos of dust (certainly not at megaparsec scales!) and so using star catalogs as foreground samples the extinction calculation offers a valuable null test. For the redMaGiC-based catalogs, star catalogs are taken from the Gaia Data Release 3 database (Gaia collaboration 2023); as before, overlapping regions are obtained by calculating the intersection of masks for each survey of interest. As the Gaia collaboration does not make a survey mask readily available, I create one using a simple function (`radec2healpixels`) described in Section 4.1.1. The result for the Gaia stars is shown in Figure 2.

Figure 2 also illustrates the gradient in the number density of stars across the DES field of view approaching the galactic plane. In theory, failing to account for this variation in source density in random catalogs might bias the resulting correlation measurement. As such, both “signal” and “random” star catalogs are independent, random subsamples of the full Gaia Data Release 3 catalog.

After all catalog joining, selections and survey masking operations, there remain:

- 169,605 galaxies in the Gaia DR3 star catalog,
- XXX galaxies in the Gaia random catalog,
- XXX galaxies in the GWSLC-2 catalog,
- and XXX galaxies in the GWSLC-2 random catalog.

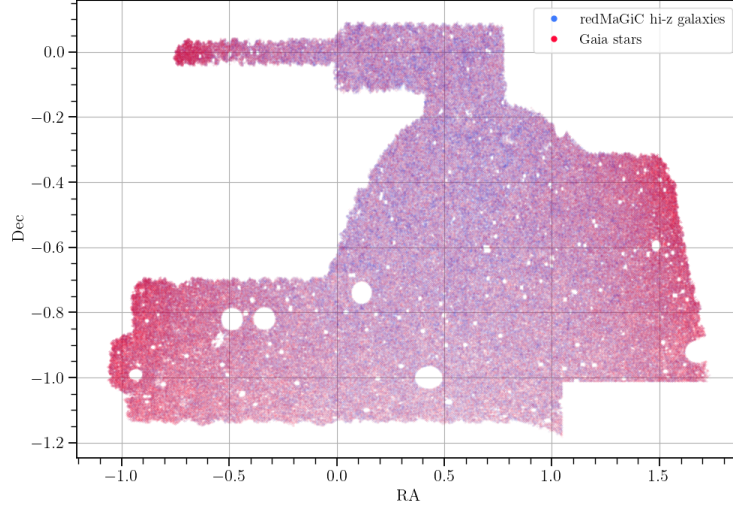


Figure 2: Sky coverage of the masked Gaia DR3 (foreground, red points) and the high-luminosity high-redshift redMaGiC galaxy catalog (background, blue points). The expected gradient in star density closer to the galactic plane is noticeable.

4 Analysis Fixes and Improvements

Since update #1, there have been several useful upgrades and bug fixes to the analysis codebase. Changes to catalog processing are summarized in Section 4.1 and changes to extinction and correlation calculations are summarized in Section 4.2.

4.1 Catalog processing updates

The major changes to catalog processing include: mitigation of a bug in random catalog generation; implementing redshift-based matching for assigning photometry to random catalogs; and creating survey masks when not pre-supplied.

4.1.1 Creation of HEALPix survey masks

Survey masks are essential for constructing catalogs matched to one another in sky coverage. Failure to properly implement proper masking for both real and random catalogs may artificially boost or suppress correlation signal. “The correlation function effectively measures the clustering relative to a random field. If the ”random” field does not have the same exclusion zones, then where the data has zeros (in the masked areas), the random field has non-zero values.”

For those cases where survey masks are not supplied for a particular catalog, I added a function to the source code called `radec2healpixels` that creates a HEALPix mask from the catalog itself by converting celestial coordinates to HEALPix pixel indices. The function can be called with a runner script, e.g.,

```
## Import function
from src.radec2healpixels import radec2healpixels

## Define variables
catname = 'dusty_halos/catalogs/gaia_stars_southern_full.fits'
```

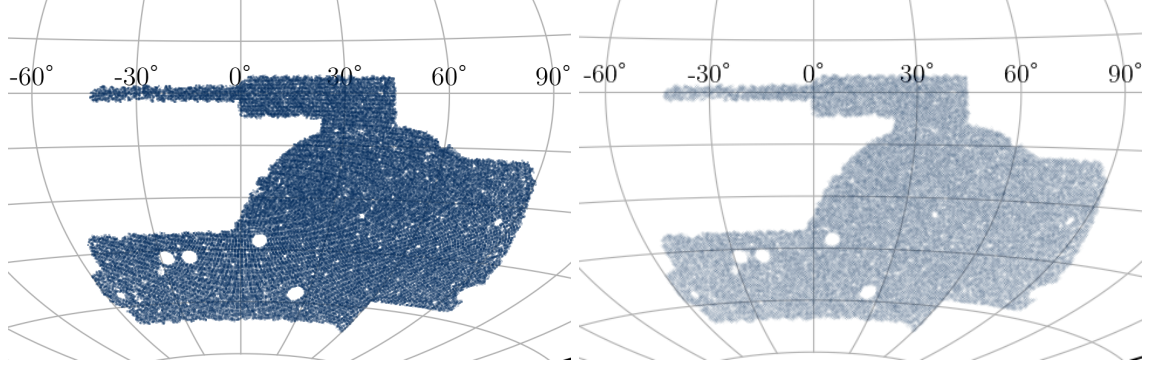


Figure 3: Sky coverage of a masked random galaxy catalog generated with two versions of FgRandom. The top panel shows the fall 2023 version, which calculated (α, δ) from randomly sampled HEALPix pixels. The bottom panel shows the result of sampling (α, δ) from a uniform distribution. The moiré pattern noticeable in the left panel has been eliminated in the right panel.

```
ra_col = 'ra'; dec_col = 'dec'
Nside = 2048

## Run
radec2healpixels(filename=catname, Nside=Nside, ra_col=ra_col, dec_col=dec_col)
```

4.1.2 Random catalog creation

As I described in the first update, for cases where random catalogs are not available, I developed a class called FgRandom (defined in `fg_random.py`). In the first release, FgRandom used a seeded random number generator and the survey mask’s HEALPix `NSIDE` parameter to randomly select HEALPix pixels and converting those to RA, Dec using the HEALPy library’s `pix2ang()` function.

An example of a random catalog generated with the original version FgRandom is shown in the top panel of Figure 3. I originally stated that the moiré effect was an artifact of projection and not present in the catalog itself. However, it never quite sat right with me, and my original statement turned out *not* to be true! It should have been obvious earlier, but

4.1.3 Redshift-matched photometry for random catalogs

The redMaGiC catalogs (both real and random) have positions and redshifts, but not magnitudes. To obtain colors for the real catalogs, I join them with their counterparts in the Y3 GOLD catalog on the `COADD_OBJECT_ID` parameter. To do the same for the random galaxy catalogs, these objects were originally assigned photometry from the Y3 GOLD catalog randomly, with no regard for redshift. Concerned that this might introduce systematic error, I implemented redshift-based matching of random galaxies. The steps of the algorithm are relatively simple:

1. Generate some number of redshift bins (I chose seventeen) and digitize the photometry and random-galaxy catalogs accordingly.
2. For each redshift bin, generate an array of random integers with as many entries as there are random galaxies in this bin and with an upper limit of the number of photometry galaxies in

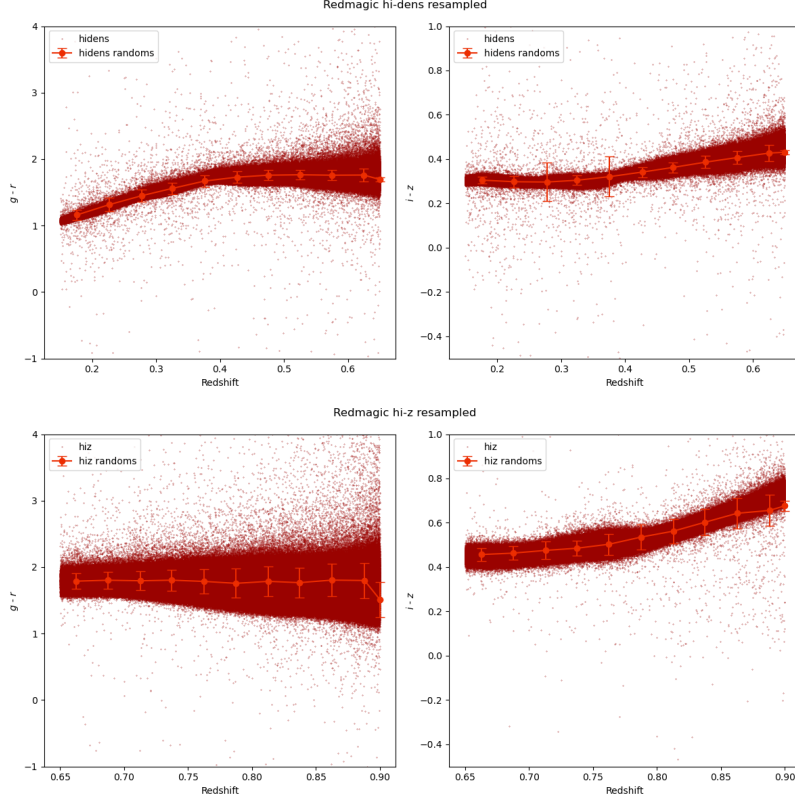


Figure 4: Color-redshift relation for the two redMaGiC catalogs (shown as small, dark red points) and the random galaxy catalogs (shown as large, bright red points connected by a solid line). The top row of figures shows the high density galaxy sample, the bottom row shows the high-luminosity high-redshift sample. In all cases, random galaxies’ colors overlap well with the real galaxy colors.

this bin.

3. Assign the g , r , i , z magnitudes of galaxies in the photometry catalog to the corresponding entries in the random-galaxy catalog elementwise.

The result is shown in Figure ?? . Note that the photometry catalogs used for this procedure are the color-augmented redMaGiC galaxy catalogs, not the full Y3 GOLD catalog. Using the full Y3 GOLD catalog produces a mismatch in colors, which is attributable to the fact that the redMaGiC galaxies are not a representative sample!

4.2 Extinction Calculation and Correlation

The code uses `TreeCorr` for the actual correlation computations and relies on a series of YAML configuration files to specify parameters, allowing for a high level of customization and reuse.

4.2.1 Extinction models

Models for dust-grain-induced reddening are generated with the `extinction` Python package²; allowed options are `calzetti00`, `fm07`, `odonnell94`, and `fitzgerald99`.

²<https://extinction.readthedocs.io/en/latest/index.html>

Filter	λ_{ref}	λ_{mean}	λ_{eff}
u	3814.33	3821.68	3856.88
g	4808.49	4862.24	4769.90
r	6417.65	6460.63	6370.44
i	7814.58	7850.77	7774.30
z	9168.85	9199.28	9154.88

Table 1: Reference wavelengths for the DECam filters used in this analysis. The u filter is included for reference only, as the Y3GOLD catalog does not provide u -band magnitudes.

The `extinction` models require that wavelengths at which to compute the extinction be specified. The Virtual Observatory’s Filter Profile Service provides reference wavelengths for the DECam filter set, which are shown in Table 1. I chose to calculate the extinction at the reference wavelength λ_{ref} of each filter, as it splits the difference between the other two measures. The `extinction` models also require values of R_V and/or A_V to be set. Values of R_V in the literature fluctuate significantly from galaxy to galaxy (possibly redshift as well), and changing R_V changes the extinction profiles. In the absence of a compelling reason to do otherwise, I used the default values of $R_V = 3.1$ (Milky Way-type dust) and $A_V = 1$ in my calculations.

Model extinctions in the DECam bandpasses are reproduced below (these are identical to Update #1):

```
wavelengths = np.array([3814.33, 4808.49, 6417.65, 7814.58, 9168.85])

# Calzetti 2000
extinction.calzetti00(wavelengths, 1, 3.1)
array([1.54676056, 1.19281617, 0.79717052, 0.54869537, 0.38008871])

# Gordon 2023
attenuation.g23(wavelengths, 1)
array([XX, XX, XX, XX, XX])
```

5 Updated Results

Figures and tables in this section summarize the results of the A_V estimation with the the two red-MaGiC galaxy catalogs and the cross-correlation with galaxy counts in WISExSCOS. The number of redshift bins for the A_V estimator calculation were chosen to balance bias and variance in the results. Five redshift slices were used for the high-luminosity, high-redshift redMaGiC sample, and fifteen redshift slices for the high-density redMaGiC sample.

Where relevant and for ease of comparison, extinction profiles calculated with a previous generation of code and the code current as of March 2024 are juxtaposed.

5.1 Extinction profiles

The principal result of this analysis, extinction profiles for the WISExSCOS-redMaGiC cross-correlations, is shown in Figure ??.

Lan & Mo (2018) and Zhu et al (2014) show that star-forming galaxies are not the only ones for which we might expect to see large halos of dust. Figure ?? implies that luminous red galaxies

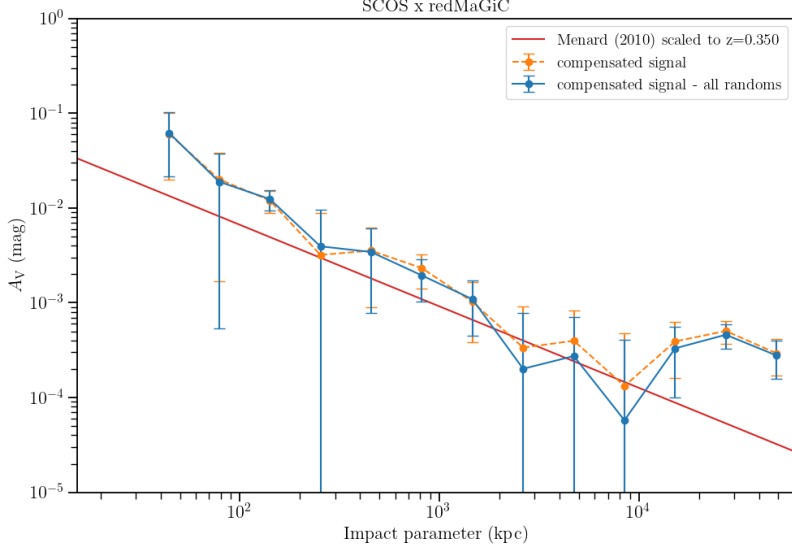


Figure 5: Extinction profile of luminous red galaxies, selected from the redMaGiC high-density sample, with $z < 0.45$. The redMaGiC high-luminosity high-redshift sample is used as a background catalog and A_V is calculated in 9 redshift bins.

also possess an extended, denser CGM halo. To test this with our own sample, I cross-correlated a low-redshift subset ($z < 0.45$) of the redMaGiC high-density sample with reddening of the high-luminosity, high-redshift sample. The resulting extinction profile is shown in Figure 5 and seems to confirm the existence of extended halos of cool, dense gas around LRGs.

As before, I explored the use of the different dust models available in the `extinction` package; A_V profiles calculated using redMaGiC high-redshift galaxies and three different dust models are shown in Figure 6. There do not appear to be any significant differences (within error bars) between the available models, suggesting that the extinction profile isn’t attributable to a fluke in a particular model.

5.2 Sanity checks and tests

A sharp change in the magnitude of the extinction at 200-400 kpc is visible in Figures ??–6; moreover, at high impact parameters the random subtraction is *adding* power to the high-density galaxy catalog. I considered the relative contributions of the different terms in the ξ_{NA_V} estimator defined in Equation 3. An example (high density redMaGiC, A_V calculated in 8 redshift bins) is shown in the left panel of Figure 8. It is apparent that the random-foreground, real-background cross-correlation is driving the excess at $r > 5$ Mpc. Without that subtraction (profile in teal and dark blue lines), the extinction falls off as one might expect. The right panel of 8 shows a similarly odd trend for the high-redshift sample: subtracting a $R_N D_{A_V}$ component makes the signal all but vanish.

My first thought was that my random foreground catalog algorithm must have a bug, but Figure 7 shows that the same $R_N D_{A_V}$ issue emerges when redMaGiC randoms are used instead. To produce Figure 7, a high-redshift “foreground” catalog at $z > 0.65$ was cross-correlated with reddening of a low-redshift “background” catalog at $z < 0.45$. In the absence of systematic error, the extinction profile should be consistent with 0. Instead, the $R_N D_{A_V}$ subtraction introduces extra signal at

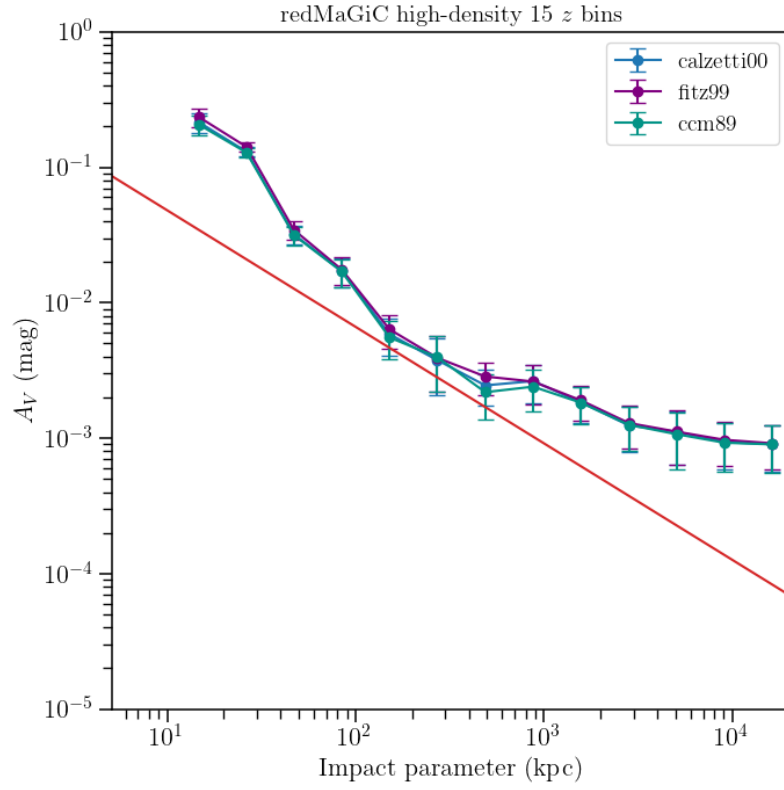


Figure 6: RedMaGiC high-redshift sample extinction profiles with A_V calculated in four redshift bins and with three different extinction models: `calzetti00` (light blue), `ccm89` (purple), and `fm07` (teal).

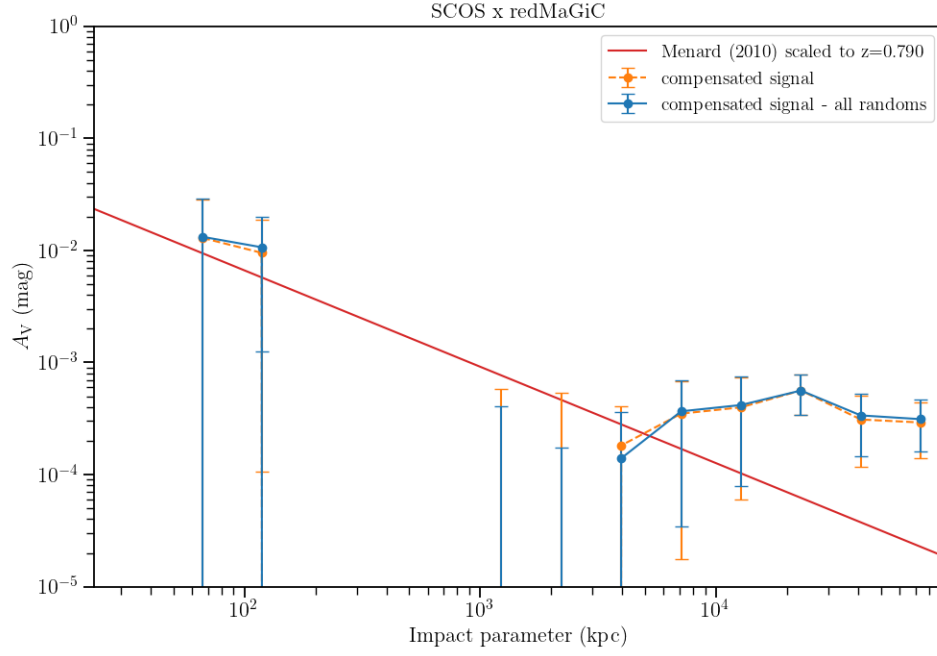


Figure 7: Null test: extinction profile resulting from cross-correlating a “foreground” sample of redMaGiC hi-z galaxies ($z > 0.65$) with reddening of actual foreground galaxies with $z < 0.45$ taken from the high-density sample. Excess around 200 kpc and at $r > 10Mpc$ are noticeable; otherwise, the curve is consistent with zero.

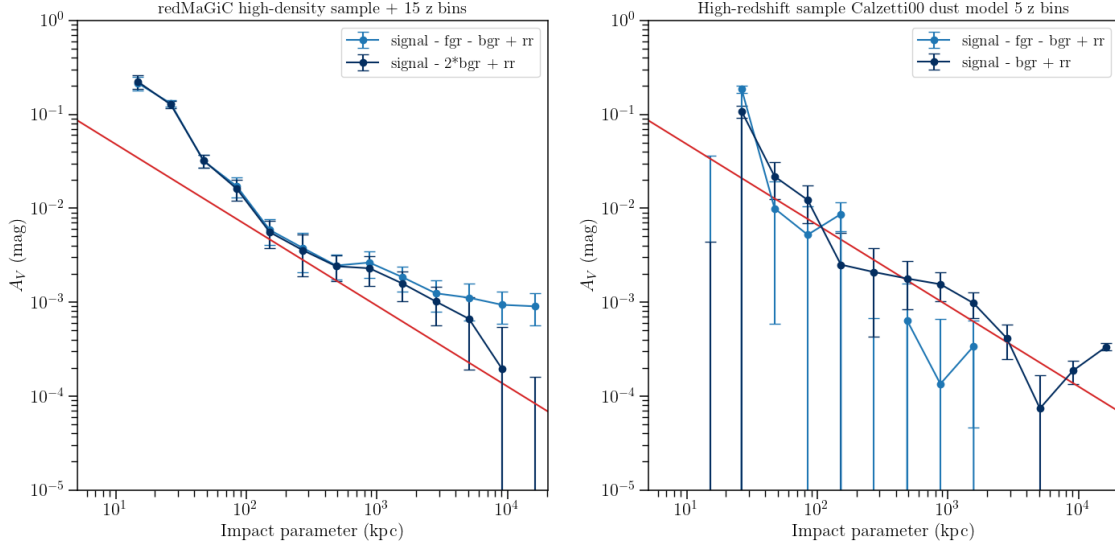


Figure 8: Extinction profiles, with and without subtraction of the random-foreground real-background cross-correlation signal. The fully random-subtracted galaxy- A_V correlations are shown in teal; the correlation without foreground random subtractions are shown in blue. An extinction profile calculated with a modified A_V estimator is shown in dark blue in the left panel.

high impact parameter. This suggested that rather than my foreground random algorithm being incorrect, our ξ_{NA_V} estimator may be flawed. Something like a Landy-Szalay estimator provided more reasonable results for the A_V profile (dark blue line, left panel of Figure 8). However, the LRG extinction profile of Figure 5 provides a counterargument to this line of reasoning: the *raw* signal displays the excess at high impact parameters, and subtraction of the $R_N D_{A_V}$ removes some of this excess. So, the cause and significance of this excess at $r > 10$ Mpc remains unclear.

6 Next steps and open questions

Before declaring victory and preparing a manuscript for submission, there are a few tasks remaining. Fundamentally, we are aiming to say something about the physics of dust and the larger galaxy environment; the following analyses could help us to that end.

1. Segregating galaxies by type (red vs. blue) informs models of differences in the larger-scale environments these galaxies inhabit. This is an easy test that we could carry out with the data we already have. Even if I don't manage to get useable results with the GWSLC-X catalog, Figure 9 shows that this could be done with the WISExSCOS catalog.
2. Evaluating $A_V < 50$ kpc as a function of star formation provides an even more direct test of the connection between galaxy disks and their halos.
3. I would like to figure out a way to get a surface dust mass as a function of radius from the galaxy. Analyzing the eBOSS galaxy sample in Lan & Mo (2018) would be an intermediate step, but it would be nice to have something more general.
4. Fitting some kind of halo model to our extinction profiles would also be cool. I don't know how to implement it, but I think it would be very illuminating to connect halo mass with dust abundance and extent.
5. Proving that our estimator can recover a known quantity of dust would be a really good cross-check of our methods. Analyzing the outputs of IllustrisTNG galaxies, which include dust, could be a way to do this.

The first two items on this list are quite simple to implement; the others are more more involved (meaning I don't know how to do them yet!). Conducting analyses 3 and 4 would provide a good excuse to reach out to our old friend Brandon Hensley, who doubtless knows the best way to go from extinction to some sort of physical model. For analysis 5, Figure ?? shows that this is at least possible in theory. I'm sure I can figure it out (or put a student on it!).

- The COSMOS catalog has blue-band magnitudes and excellent redshifts; the sky area (and thus sample size) is relatively small, but building foreground and background catalogs from COSMOS could be worth the effort, as the blue bandpasses are especially sensitive to dust extinction.
- Using the GALEX-SDSS-WISE Legacy Catalog as a foreground has the advantage of providing star formation rates, enabling an examination of extinction as a function of SFR.
- Using the DES MAGLIM catalog would vastly increase the number of galaxies available for reddening analysis. I could then reevaluate some of the odd trends evident in the high-redshift redMaGiC sample, namely, a extinction profile that disappears as A_V becomes less biased.

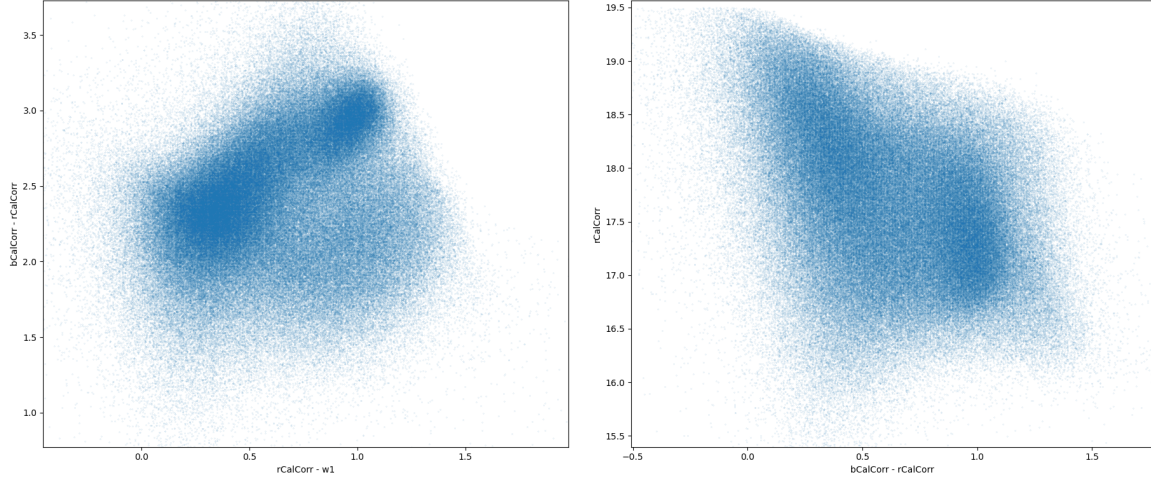


Figure 9: Color-color (left) and color-magnitude (right) plots for the WISExSCOS catalog galaxies used in this analysis. The left panel shows $bCalCorr - rCalCorr$ versus $rCalCorr - W1$; the right panel shows $rCalCorr$ versus $bCalCorr - rCalCorr$. Distinct loci of blue and red galaxies are discernible in both panels.

- The most powerful alternative catalog would be the eBOSS SDSS catalog, which has millions of galaxies, lots of blue photometry, and would allow us to reproduce the spectroscopy-based Lan & Mo result in extinction.

Before too much time passes, we should also strategize about what to put in the paper³! Eric mentioned UV studies, and Jim had been planning a dust emission study for a while. Are either of these still in progress?

7 Bonus: updates from Aspen dust workshop

My talk at “The Physics and Impact of Astrophysical Dust: from Star Formation through Cosmology” Aspen Center for Physics Winter Conference went really well! For your reference, the slides are attached. I got some good questions and suggestions, including to update the dust extinction law in the A_V calculations (see Section ??). I learned about current hydrodynamical simulation efforts to understand the entrainment and survival of dust in the IGM – specifically the work of Paul Torrey’s group. Some good references are here <https://academic.oup.com/mnras/article/503/1/336/6134059> and here <https://arxiv.org/abs/2403.03711>. He offered to give some theoretical comment on our findings when we are ready to publish, which is great! The conference also offered the opportunity to reconnect with Brandon Hensley. This was definitely one of the more scientifically productive (and pleasant) conferences I have attended.

³I am in favor of publishing a few smaller papers rather than one 50-page behemoth.